

HOMEWORK ASSIGNMENT 8

ENTROPY AND THE GIBBS RELATIONS

1. (10 points) Fill out the Exam 1 exam wrapper, which will be available on Canvas.
2. (20 points) A reversible Carnot cycle operates between a high-temperature reservoir of 800 K and a low-temperature reservoir of 300 K. If the heat rejected from the cycle is equal to 10 kJ, calculate the following:
 - a. The thermal efficiency of the cycle
 - b. The change in entropy of the system during heat addition
 - c. The change in entropy of the system during heat rejection
 - d. The net work of the cycle
3. (16 points) Calculate the change in mass-specific entropy in the following situations. Identify which assumptions you use for each of the calculations. Use the following properties for air: $R=287 \text{ J/kg-K}$ and $c_v = 720 \text{ J/kg-K}$.
 - a. Isothermal compression of air from $10 \text{ m}^3/\text{kg}$ to $4 \text{ m}^3/\text{kg}$
 - b. Isothermal compression of air from 0.1 MPa to 1.7 MPa
 - c. Isobaric heating of air from 300 K to 1200 K
 - d. Isobaric heating of water at 1 MPa from a saturated liquid to a saturated vapor
4. (20 points) Calculate the final state point for isentropic processes in air, where $\gamma=1.4$.
 - a. Find P_2 for $v_1=0.2 \text{ m}^3/\text{kg}$, $P_1=0.1 \text{ MPa}$, $v_2=0.03 \text{ m}^3/\text{kg}$
 - b. Find v_2 for $v_1=0.1 \text{ m}^3/\text{kg}$, $T_1=400 \text{ K}$, $T_2=500 \text{ K}$
 - c. Find T_2 for $P_1=0.1 \text{ MPa}$, $T_1=300 \text{ K}$, $P_2 = 1 \text{ MPa}$
 - d. Find v_2 for $P_1= 5 \text{ MPa}$, $v_1=0.5 \text{ m}^3/\text{kg}$, $P_2=10 \text{ MPa}$
 - e. Find P_2 for $T_1=400 \text{ K}$, $P_1=2 \text{ MPa}$, $T_2=750 \text{ K}$
5. (14 points) Steam expands isentropically in a piston-cylinder arrangement from a pressure of $P_1 = 2 \text{ MPa}$ and a temperature of $T_1 = 500 \text{ K}$ to a saturated vapor at State 2.
 - a. Draw this process on a T-S diagram.
 - b. Calculate the mass-specific entropy at State 1.
 - c. What is the mass-specific entropy at State 2?
 - d. Calculate the pressure and temperature at State 2.
6. (20 points) Before starting at Penn State, I worked at Sandia National Laboratories in an engine lab where we studied combustion processes in diesel engines. I worked in a lab with a single-cylinder engine that had a piston made of optical-grade quartz so that we could see inside the combustion chamber – it was really cool! One of the things we could not do was measure the temperature at the top of the compression stroke – regular temperature measurements aren't that fast. Instead we had to calculate it. Our engine had a compression ratio of $V_1/V_2=11.2$ (this is low for a diesel engine – it's a result of all the changes that we had to make to the engine to make it optically accessible). Because the low compression ratio, we had to boost the incoming air to get the right thermodynamic conditions at the top of the compression stroke to match what we'd see in real engines. In this problem, you're going to calculate the temperature and pressure at the

top of an isentropic compression in this engine for a range of operating conditions – assume the ratio of specific heats of air is 1.4.

- a. Use Excel or Matlab to plot pressure after an isentropic compression for $T_1=320$ K as a function of initial pressure for $P_1 = 0.1$ MPa to 0.2 MPa.
- b. Use Excel or Matlab to plot temperature after an isentropic compression for $P_1=0.15$ MPa and a range of temperatures from $T_1=280$ K to 350 K.
- c. Read a bit about the work we did in the lab here:
<https://www.energy.gov/eere/vehicles/advanced-combustion-strategies> The video on the top of the page was taken in the engine I worked in. Discuss at least one strategy for reducing engine emissions in diesel engines. How could the thermodynamic condition at the top of a compression stroke, right before combustion, change emissions?